Order of Growth

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Let f(n) and g(n) be the time taken by two algorithms where n >= 9 and f(n) and g(n) are also greater than equal to 0. A function f(n) is said to be growing faster than g(n) if g(n)/f(n) for n tends to infinity is 0 (or f(n)/g(n) for n tends to infinity is infinity).

> ***Example 1:*** *f(n) = 1000, g(n) = n + 1*
> *For n > 999, g(n) would always be greater than f(n) because order of growth of g(n) is more than f(n).*
>
> ***Example 2****: f(n) = $4n^2$, g(n) = 2n + 2000*
> *f(n) has higher order of growth as it grows quadratically in terms of input size.*

How do we Quickly find order of Growth?

When n >= 0, f(n) >= 0 and g(n) >= 0, we can use the below steps.

- Ignore the order terms.
- Ignore the constants

For example,

> ***Example 1 :*** *$4n^2$ + 3n + 100*
> *After ignoring lower order terms, we get*
> *$4n^2$*
> *After ignoring constants, we get*
> *n^2*
> *Hence order of growth is n^2*
>
> ***Example 1 :*** *100 n Log n + 3n + 100 Log n + 2*
> *After ignoring lower order terms, we get*
> *100 n Log n*
> *After ignoring constants, we get*
> *n Log n*
> *Hence order of growth is n Log n*

How do we compare two order of growths?

The following are some standard terms that we must remember for comparison.

$c < Log\ Log\ n < Log\ n < n^{1/3} < n^{1/2} < n < n\ Log\ n < n^2 < n^2\ Log\ n < n^3 < n^4 < 2^n < n^n$

Here c is a constant

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- Asymptotic Analysis
- Worst, Average and Best Cases of Algorithms
- Asymptotic Notations

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